

Capping the Fixed Set

The P -axis of the split-source decomposition, once q may be fixed first

Resolves. *Decomposition for Split Unpredictable Sources* (c/0010). **Partially, and for a relaxed statement.** The conjecture’s hypothesis on P is removed entirely, on the region (H2), provided the mixture family is allowed to depend on the observer’s query budget q . That is a relaxation of the conjecture, not the conjecture. The order-respecting statement, with an absolute constant and a q -free family, is *not* settled here.

Status. Post-audit. Every result below has been through six independent blind referee passes spanning two model families; five judged the artifact and all five returned DEFECTS, none of them computational. The mathematics survived; seven claims *about* the mathematics did not, and are corrected here. The revision carrying these corrections has itself had no pass.

Reader’s guide

The idea, in one sentence. The conjecture asks for *some* family of P -mixtures; it never asks that the fixed sets be large, and Definition 1 bounds them by $|I| \leq P$ rather than fixing $|I|$. So when the requested P exceeds what the proof can afford, do not use it — build the family at the balance point instead. A P_0 -mixture with $P_0 \leq P$ is a P -mixture.

What that buys. The hypothesis (H1) of the campaign’s Theorem E’, which caps P at roughly $\sqrt{\sigma' q^+ / \delta}$, disappears. The cost is the constant C , which moves from 8 to 13; the constant c does not move.

What it does not buy. Two things, and both were overclaimed in the first draft of this work before referees caught them.

- The family must be allowed to depend on q . The Contract’s Remark `rem:order` forbids that: the family is chosen after P and γ but *before* q . So Theorem H proves a relaxation, and the window the campaign records as open — absolute C , q -free family — stays open.
- Keeping the family q -free costs a factor $\sqrt{q^+}$ in the constant (Theorem H’). It is tempting to conclude that the $\sqrt{q^+}$ therefore *is* the price of q -independence. That does not follow, and it is not claimed here: what is shown is that q -dependence *suffices* to remove the factor within this route. No lower bound against q -free families exists anywhere in the campaign.

What remains. The hypothesis (H2), a condition on the output size M , is carried unchanged through everything below. It is the whole residue on the P -axis, and §4 is about what it means and whether applications satisfy it. They do for key extraction and do not for long-output extraction.

1 Setting

Fix $N, M \geq 2$ and let $\text{Fun} := \{f : [N] \times [N] \rightarrow [M]\}$, with $H \leftarrow_{\$} \text{Fun}$. A pair (S_1, S_2) of sources is *split* when the two use independent coins and neither sees the other's output; each reads all of H and outputs a point $x_i \in [N]$ together with leakage z_i with $|z_i| \leq \sigma_i$. The pair is δ -*unpredictable* when no predictor given H and the whole leakage vector guesses either point with probability more than δ . All algorithms are computationally unbounded; only queries count.

Throughout, with logarithms to base 2,

$$\begin{aligned} \sigma &:= \sigma_1 + \sigma_2, & \sigma' &:= \sigma + 2 \log N, & q^+ &:= q + 1, & S &:= \sqrt{\sigma' q^+ \delta}, \\ t_q &:= \sqrt{\sigma' q^+ / \delta}, & t_0 &:= \sqrt{\sigma' / \delta}, & t_q &= t_0 \sqrt{q^+}, & B &:= \sigma' + \log \gamma^{-1}. \end{aligned}$$

Note $\delta \geq 1/N$ always, since a uniform guess predicts x_i with probability $1/N$; and $\sigma' \geq 2$, which needs $N \geq 2$ and is used three times below.

Definition 1 (bit-fixing sources, and consistency). A distribution Y over Fun is *P-bit-fixing* if there are a set $I \subseteq [N] \times [N]$ with $|I| \leq P$ and a map $a : I \rightarrow [M]$ such that a draw $g \leftarrow Y$ satisfies $g(u) = a(u)$ for every $u \in I$ and is uniform on $[M]$ and independent across the points $u \notin I$. A *P-mixture* is a finite indexed family of *P-bit-fixing* components; it is *consistent with f* when $a(u) = f(u)$ on I for every component.

The inequality $|I| \leq P$ is the whole hinge, and it is worth pausing on. It is not a stylistic choice: the same convention appears verbatim in Coretti–Dodis–Guo–Steinberger, whose Definition 1 reads “*P-bit-fixing* if it is fixed on *at most P* coordinates”.

2 The construction

Given a requested $P \in \mathbb{N}$ and $\gamma \in (0, 1)$, put

$$\gamma_0 := \max(\gamma, N^{-2}), \quad P_0 := \min(P, \lceil t_q \rceil), \quad \mathcal{Y} := \mathcal{Y}^{P_0, \gamma_0/2},$$

where $\mathcal{Y}^{\cdot}$ is the family the campaign's Theorem E'' takes as given. Two observations license this.

Lemma H0 (admissibility). Every member of $\mathcal{Y}$ is a P -mixture consistent with its index; $\mathcal{Y}$ is indexed by (f, ζ) and by nothing else; and Theorem E'' applies to it at (P_0, γ_0) .

Proof. Each member is a P_0 -mixture, so each component has fixed set I with $|I| \leq P_0 \leq P$. Definition 1 asks $|I| \leq P$ and nothing more, so the same witness (I, a) that proves P_0 -bit-fixing proves P -bit-fixing. The uniformity and independence conditions off I refer to the same unchanged I , and consistency is a property of the values a , so both survive verbatim. The index set is unchanged, as rem:index requires. For Theorem E'': $\gamma_0 \in (0, 1)$ since $\gamma < 1$ and $N^{-2} \leq 1/4$; hypothesis (H1) reads $P_0 \leq t_q + 1$ and holds because $P_0 \leq \lceil t_q \rceil \leq t_q + 1$; and (H2) does not mention P , so it is unaffected. $\square$

Lemma H1 (three slack bounds). $\log \gamma_0^{-1} \leq \log \gamma^{-1}$; $\log \gamma_0^{-1} \leq \sigma'$, hence $\sigma' + \log \gamma_0^{-1} \leq 2\sigma'$; and $N^{-2} \leq \frac{1}{4}S$.

Proof. $\gamma_0 \geq \gamma$ gives the first, and it is what keeps c at 2: without it the clamp would inflate the leading term rather than the additive one. $\gamma_0 \geq N^{-2}$ gives $\log \gamma_0^{-1} \leq 2 \log N \leq \sigma'$. For the third, $\delta \geq 1/N$, $\sigma' \geq 2$ and $q^+ \geq 1$ give $S \geq \sqrt{2/N}$, so $N^{-2} \leq \frac{1}{4}S \iff N^{-3} \leq \frac{1}{2}$, true for every $N \geq 2$. $\square$

3 The two theorems

Theorem H (at fixed q , no hypothesis on P). Let $N, M \geq 2$, fix $q \in \mathbb{N} \cup \{0\}$ and suppose (H2). For every $P \in \mathbb{N}$ and every $\gamma \in (0, 1)$, the family $\mathcal{Y}$ above consists of P -mixtures consistent with their indices, depends only on (S_1, S_2, P, γ, q) , and satisfies, for every q -query challenge-oblivious observer D and with no restriction on challenge resolution,

$$\text{Adv}_{\mathcal{Y}, D} \leq \frac{2(\sigma' + \log \gamma^{-1}) q^+}{P} + 13S + \gamma.$$

Proof. Lemma H0 licenses Theorem E'' at (P_0, γ_0) , giving $\text{Adv} \leq 2(\sigma' + \log \gamma_0^{-1})q^+/P_0 + 8S + \gamma_0$. Two cases.

Case (a), $P \leq \lceil t_q \rceil$. Then $P_0 = P$, and $\log \gamma_0^{-1} \leq \log \gamma^{-1}$ makes the first term at most $2Bq^+/P$, the target's own. With $\gamma_0 \leq \gamma + \frac{1}{4}S$ the total is $2Bq^+/P + 8\frac{1}{4}S + \gamma$.

Case (b), $P > \lceil t_q \rceil$. Then $P_0 = \lceil t_q \rceil \geq t_q$, and $\sigma' + \log \gamma_0^{-1} \leq 2\sigma'$ gives

$$\frac{2(\sigma' + \log \gamma_0^{-1})q^+}{P_0} \leq \frac{4\sigma' q^+}{t_q} = 4\sigma' q^+ \sqrt{\frac{\delta}{\sigma' q^+}} = 4S,$$

so the total is at most $12\frac{1}{4}S + \gamma$, and the target's first term is non-negative.

Both cases are dominated by $2Bq^+/P + 13S + \gamma$. $\square$

This is the Contract's conjectural *display* at $c = 2$, $C = 13$, with no hypothesis on P . It is not the conjecture: that requires the family to depend only on (S_1, S_2, P, γ) , and a q -dependent family satisfies a relaxation of it. Of the five extra units in C , four are case (b)'s capping and a quarter is the γ -clamp; three quarters go unused.

Theorem H' (one q -free family, every P , at a cost of $\sqrt{q^+}$). Let $N, M \geq 2$ and suppose (H2) holds at *every* $q \in \mathbb{N} \cup \{0\}$ — equivalently $M \leq \sigma'/(2\delta)$, the intersection binding at $q = 1$. Then the family $\mathcal{Y}' := \mathcal{Y}^{P'_0, \gamma_0/2}$ with $P'_0 := \min(P, \lceil t_0 \rceil)$ depends only on (S_1, S_2, P, γ) — not on q — and satisfies, for every $P \in \mathbb{N}$, every $q \in \mathbb{N} \cup \{0\}$ and every q -query challenge-oblivious observer D ,

$$\text{Adv}_{\mathcal{Y}', D} \leq \frac{2(\sigma' + \log \gamma^{-1}) q^+}{P} + (4\sqrt{q^+} + 9)S + \gamma.$$

Proof. As in Lemma H0, with (H1) at P'_0 holding for every q at once, since $P'_0 \leq \lceil t_0 \rceil \leq t_0 + 1 \leq t_q + 1$ by $t_0 \leq t_q$. The hypothesis (H2) is *not* uniform in q and is supplied at every q by assumption. Case (a) is as before. Case (b), $P > \lceil t_0 \rceil$: now $P'_0 \geq t_0$, so $4\sigma' q^+/t_0 = 4q^+ \sqrt{\sigma' \delta} = 4\sqrt{q^+} \cdot S$, using $S = \sqrt{q^+} \sqrt{\sigma' \delta}$. $\square$

Remark 1 (the hypothesis of Theorem H', and a defect worth recording). The first draft of this theorem wrote “suppose (H2) holds at q ” and then quantified its conclusion over *every* q — leaving q free in the hypothesis and rebinding it in the conclusion. Read literally the hypothesis is *vacuously* satisfiable at $q = 0$, where the relevant quantity is 0, so the theorem would have delivered the conjecture for every M and contradicted its own statement of what remains open. A non-vacuous witness that the readings differ: $N = 2^{12}$, $\sigma = 0$, $\delta = 2^{-12}$, $M = 2^{16}$. There $8S < 1$ at both $q = 0$ and $q = 1$, (H2) holds at $q = 0$, and at $q = 1$ it fails on both arms. Three of the six referees raised this.

4 What (H2) is, and whether applications satisfy it

(H2) asks that the residue of the campaign’s extraction bound fit inside one unit of the budget S :

$$\min\{\mu'(\min(qM, N^2)), 2\delta\sqrt{M}\} \leq S, \quad \mu'(s) := \min(s\delta, 2(s\delta^2)^{1/3}, 1).$$

It is not a condition about decomposition at all. It is inherited from routing the decomposition through extraction, and it enters at exactly one step.

The second arm has a clean reading. A random function’s image distribution $\nu(v) = |f^{-1}(v)|/N^2$ is lumpy: throwing the pair’s $\approx 1/\delta^2$ effective probability mass into M buckets gives a balls-in-bins deviation $\approx \sqrt{M\delta^2} = \delta\sqrt{M}$. The real challenge is drawn from that lumpy law and the decomposed one is uniform off the fixed set, so the lumpiness genuinely appears in the advantage. Exploiting it, however, requires knowing *which* values are over-represented, and that means learning f . That is why $q = 0$ is free for every M , and why the only known lower bound witnessing $\delta\sqrt{M}$ spends N^2 queries.

When D has full challenge resolution — as a general distinguisher $D^H(y, \mathbf{z})$ does, since it may steer its queries on y — the second arm is the smaller for $q \geq 1$, $M \geq 4$, and (H2) becomes, in bits,

$$\log M \leq k + \log(\sigma' q^+) - 2, \quad k := \log(1/\delta).$$

Do not ask for more output bits than one source’s unpredictability, plus a logarithmic allowance. The q -dependence is only additive-logarithmic: q from 2^0 to 2^{128} moves the allowance by 127 bits. So the corner where (H2) binds is not really about query budgets at all — it is about asking for a long output.

The two failing rows differ in kind and only one matters. Row 4 fails (H2), but the bound there is 2^{-27} anyway, so nothing is lost that one had. Rows 5 and 6 are the real gap: security is still 2^{-91} and (H2) blocks the statement purely for asking for a long output. That is precisely the sponge and Merkle–Damgård streaming use which motivated wanting a decomposition route in the first place.

Remark 2 (the evidence for (H2) does not reach the corner that binds). The only lower bound in the campaign exhibits an observer making N^2 queries, and κ is monotone in q — a q -query observer may ignore queries — so a lower bound at $q = N^2$ constrains $\kappa(1)$ not at all. Meanwhile (H2) is free at $q = N^2$, permitting

setting	k	output	allowed	(H2)	bound
128-bit key, 256-bit-unpredictable sources	256	128	≤ 328	yes	2^{-91}
256-bit key, 256-bit-unpredictable sources	256	256	≤ 328	yes	2^{-91}
128-bit key, 128-bit-unpredictable sources	128	128	≤ 200	yes	2^{-27}
256-bit key, 128-bit-unpredictable sources	128	256	≤ 200	no	2^{-27}
512-bit output, 256-bit sources	256	512	≤ 328	no	2^{-91}
1 Mbit stream, 256-bit sources	256	2^{20}	≤ 328	no	2^{-91}

Table 1: (H2) at application parameters, with $\sigma = 64$, $q = 2^{64}$, $\log N = 512$. Computed by `c/0010/campaign/checks/h2-applications.py`.

$M \leq \sigma' N^2 / (4\delta)$, and binds at small q . *No artifact in this campaign gives any evidence that the extraction advantage carries M -dependence where (H2) actually bites.* The sharp question is therefore narrower than the campaign has been recording: does $\kappa(q)$ have any M -dependence for small q ? A refuter pass hunting a small- q , large- M distinguisher is the cheap direction, and the campaign has never run one.

5 What is settled, and what is not

Settled, on the region (H2). The P -axis, at fixed q , in full: no hypothesis on P survives in Theorem H.

Not settled.

- *The conjecture as stated.* Theorem H's family depends on q ; Theorem H's constant does. Neither is the conjecture, which wants both a q -free family and an absolute C . The window the campaign records as open is exactly as open as before, only better described: the residue on the P -axis is a factor $\sqrt{q^+}$ in the constant under q -free families, on $M \leq \sigma' / (2\delta)$.
- *The M -corner.* (H2) is carried, not discharged, except at $q = 0$ where it is vacuous for every M .
- *Whether the $\sqrt{q^+}$ is necessary.* Only achievability is shown.

A redirection. The campaign's deepest effort goes into removing the $P\delta$ term so that a *large* fixed set can be afforded. At fixed q that work is unnecessary: one never needs a fixed set larger than $\lceil t_q \rceil$. That programme is aimed solely at the q -free statement, where the cap must sit at t_0 and the $\sqrt{q^+}$ is real.

6 Verification status

This work has been through six independent blind referee passes in fresh context, spanning two model families: five on the artifact (lenses: the definitional step; arithmetic and constants; quantifier order; case coverage and degenerate instances; citation fidelity) and one on three proposed simplifications. All five artifact passes returned DEFECTS.

Confirmed after adversarial attempt. Lemma H0 was attacked by four referees independently, each trying to construct a reading of Definition 1 on which it fails; none could. The arithmetic referee re-derived every inequality by hand, reproduced the machine check’s grid sizes exactly including collision counts, found *no computational error*, and confirmed that $c = 2$ and $C = 13$ follow from the displayed inequalities rather than from the grid. The case analysis is exhaustive with no boundary gap. No referee found a violation of the index-set restriction, or a leak of D , $\mathbf{x}$ or the challenge into either family.

Corrected. Seven class-(A) and class-(C) findings, every one of them a claim *about* the results rather than a result: Theorem H’'s malformed hypothesis; an identification of the $\sqrt{q^+}$ with the Contract’s own separation that conflated a threshold in P with a degradation of C and overrode a documented caution; a claim to reproduce a construction whose parameter $A := \sigma + 2 + \log(1/\gamma)$ is not σ' ; a machine check described as reaching $q \leq 10^7$ whose vacuity filter in fact admitted nothing above $q^+ = 410$, so that $\sqrt{q^+}$ was exercised only to $\sqrt{11}$; the unstated standing hypothesis $N, M \geq 2$; and two summary lines scoped to a stronger statement than the body proves.

Still owed. The revision carrying these corrections has had *no* pass. Everything here is a corollary of the campaign’s Theorem E’', which rests on a corollary sitting in an artifact with zero verification passes, whose only referee objection was overruled and never retested. The audit verified this work *given* its dependencies; it did not touch them.